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Senergy CANBUS Protocol of High Voltage System

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Revision History

**The version is denoted with 1.0, or 1.1 or 1.2...*

Item	Version	Description	Prepared By	Date
1	1.0	Initial version	Ellie	2023-05-22

CANBUS frame format:

Both 29 bits identifier (extended frame format) and 11 bits identifier (standard frame format) are used to transfer, and the bus transmission rate is 500kbps.

The host device sends the 29-bit ID extended frame， the battery respond the 29-bit extended frame data， The host The host device sends the 11-bit ID standard frame， the battery also respond the 11-bit standard frame data.

Communication Mode:

After the host device sends the check or control command, the battery system responds data.

Report Sending is LSB.

The parts of this protocol in gray font are reserved for specific content.

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1 Query Instruction

1.1 Ensemble information

● Host device sending data:

CAN ID：0x4200(29 bits identifier) or 0x420(11 bits identifier) (this is Broadcast Frame)

Byte0	0	Ensemble Information
Byte1	Reserve	
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

● The battery pile device responds the following data:

CAN ID：0x4210+Addr(29 bits identifier)(Equipment Address: Addr. = 0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x421 (11 bits identifier)

Byte0	Battery Pile Total Voltage	Resolution: 0.1V Offset: 0
Byte1		
Byte2	Battery Pile Current	Resolution: 0.1A Offset: -3000A
Byte3		
Byte4	second level BMS Temperature	Resolution: 0.1 ºC Offset: -100 ºC
Byte5		
Byte6	SOC	Resolution: 1%, Offset: 0
Byte7	SOH	Resolution: 1%, Offset: 0

CAN ID：0x4220+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x422 (11 bits identifier)

Byte0	Charge Cutoff Voltage	Resolution: 0.1V Offset: 0
Byte1		
Byte2	Discharge Cutoff Voltage	Resolution: 0.1V Offset: 0
Byte3		
Byte4	MAX Charge Current	Resolution: 0.1A Offset: -3000A
Byte5		
Byte6	MAX Discharge Current	Resolution: 0.1A Offset: -3000A
Byte7		

CAN ID：0x4230+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

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CAN ID: 0x423 (11 bits identifier)

Byte0	MAX Single Battery Cell Voltage	Resolution: 0.001V Offset: 0
Byte1		
Byte2	MIN Single Battery Cell Voltage	Resolution: 0.001V Offset: 0
Byte3		
Byte4	MAX Single Battery Cell Voltage Number	Resolution: 1 Offset: 0
Byte5		
Byte6	MIN Single Battery Cell Voltage Number	Resolution: 1 Offset: 0
Byte7		

CAN ID：0x4240+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x424(11 bits identifier)

Byte0	MAX Single Battery Cell Temperature	Resolution: 0.1 ºC Offset: -100
Byte1		
Byte2	MIN Single Battery Cell Temperature	Resolution: 0.1 ºC Offset: -100
Byte3		
Byte4	MAX Single Battery Cell Temperature Number	Resolution: 1 Offset: 0
Byte5		
Byte6	MIN Single Battery Cell Temperature Number	Resolution: 1 Offset: 0
Byte7		

CAN ID：0x4250+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x425(11 bits identifier)

Byte0	Basic Status	See Table 1 for details.
Byte1	Reserve(Cycle Period)	
Byte2		
Byte3	Error	See Table 2 for details.
Byte4	Alarm	See Table 3 for details.
Byte5		
Byte6	Protection	See Table 4 for details.
Byte7		

Table 1: Basic Status

Bit7	Reverse
Bit6	Reverse
Bit5	Reverse
Bit4	0：Null; 1：Balance charge request
Bit3	0：Null; 1：Forced charge request
Bit2	0: Sleep, 1: Charge, 2: Discharge, 3: Idle, 4~7: Reserve
Bit1	
Bit0	

Table 2: Fault

Bit7	Other error (See Table 5&6)
Bit6	Battery cell error
Bit5	RELAY_ERR/ Relay Check Error
Bit4	RV_ERR/ Input transposition Error
Bit3	DCOV_ERR/ Input Over Voltage Error
Bit2	IN_COMM_ERR/ Internal Communication Error
Bit1	TMPR_ERR/ Temperature Sensor Error
Bit0	VOLT_ERR/ Voltage Sensor Error

Table 3: Alarm

Bit15	Other Alarm (See Table 7)
Bit14	Current leakage alarm
Bit13	Fan Alarm
Bit12	Terminal High Temperature Alarm
Bit11	Reserve(MHV: Module High Voltage Alarm)
Bit10	MLV: Module Low Voltage Alarm
Bit9	DOCA: Discharge Over Current Alarm
Bit8	COCA: Charge Over Current Alarm
Bit7	DHT: Discharge Cell High Temperature Alarm
Bit6	DLT: Discharge Cell Low Temperature Alarm
Bit5	CHT: Charge Cell High Temperature Alarm
Bit4	CLT: Charge Cell Low Temperature Alarm
Bit3	PHV: Charge system High Voltage Alarm
Bit2	PLV: Discharge system Low Voltage Alarm
Bit1	BHV: Single Cell High Voltage Alarm
Bit0	BLV: Single Cell Low Voltage Alarm

Table 4: Protection

Bit15	Reserve
Bit14	Reverse
Bit13	Reverse
Bit12	Reserve(BUV2: Module 2nd level under voltage protection)
Bit11	MOV: Module Over Voltage Protect
Bit10	MUV: Module Under Voltage Protect
Bit9	DOC: Discharge Over Current Protect
Bit8	COC: Charge Over Current Protect
Bit7	DOT: Discharge Cell Over Temperature Protect
Bit6	DUT: Discharge Cell Under Temperature Protect
Bit5	COT: Charge Cell Over Temperature Protect
Bit4	CUT: Charge Cell Under Temperature Protect
Bit3	POV: Charge system Over Voltage Protect
Bit2	PUV: Discharge system Under Voltage Protect
Bit1	BOV: Single Cell Over Voltage Protect

Bit0	BUV: Single Cell Under Voltage Protect

CAN ID：0x4260+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x426 (11 bits identifier)

Byte0	Module Max. Voltage	Resolution: 0.001V Offset: 0
Byte1		
Byte2	Module Min. Voltage	Resolution: 0.001V Offset: 0
Byte3		
Byte4	Module Max. Voltage Number	Resolution: 1 Offset: 0
Byte5		
Byte6	Module Min. Voltage Number	Resolution: 1 Offset: 0
Byte7		

CAN ID：0x4270+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x427 (11 bits identifier)

Byte0	Module Max. Temperature	Resolution: 0.1 ºC Offset: -100
Byte1		
Byte2	Module Min. Temperature	Resolution: 0.1 ºCe Offset: -100
Byte3		
Byte4	Module Max. Temperature Number	Resolution: 1 Offset: 0
Byte5		
Byte6	Module Min. Temperature Number	Resolution: 1 Offset: 0
Byte7		

CAN ID：0x4280+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x428 (11 bits identifier)

Byte0	Charge forbidden mark	0xAA：effect; other value: NULL
Byte1	Discharge forbidden mark	0xAA：effect; other value: NULL
Byte2	Reserve(heart beat signal)	0x00~0xFF cycle, increasing by 1 at each answer
Byte3	Reserve(SOE / State of Energy)	Resolution: 1%, Offset: 0, Percentage of discharged energy
Byte4	Reserve(System operation status)	Only valid with special firmware of MBMS. Linked with section 2.4. 0x11:Standby(self-inspection is finished but relay didn't close, request to command the system at 0x1094 to close relay, the status will then switch to 'Run') 0x22:Run(close relay)

Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

CAN ID：0x4290+Addr(29 bits identifier)(Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x429 (11 bits identifier)

Byte0	Reserve（System error list Extension 1）	See Table 5 for details.
Byte1	Reserve（System error list Extension 2）	See Table 6 for details
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve（Alarm status extension 1）	See Table 7 for details
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

Table 5: System error list Extension 1

Bit7	Self-inspection Module Serial Amount Error
Bit6	Emergency stop
Bit5	Insulation fault
Bit4	Chip error
Bit3	Self-inspection volt error
Bit2	Internal bus error
Bit1	BMIC error
Bit0	Shutdown circuit error

Table 6: System error list Extension 2

Bit7	Current IC Error
Bit6	Current leakage error
Bit5	All BMS offline error
Bit4	Reserve
Bit3	Module internal I2C error
Bit2	Communication error between string BMS and MBMS
Bit1	Self-inspection Initialize Error
Bit0	Self-inspection module capacity Error

Table 7: Alarm status Extension 1

Bit7	Reserve
Bit6	Reserve
Bit5	Reserve
Bit4	Reserve
Bit3	String BMS self-disconnect alarm
Bit2	String BMS communication offline alarm
Bit1	Cell temperature imbalance alarm

Bit0	Cell voltage imbalance alarm

1.2 system equipment information

●The host device sends the data:

CAN ID：0x4200 (29 bits identifier) or 0x420 (11 bits identifier) (is broadcast frame)

Byte0	2	2 is system equipment information
Byte1	Reserve	
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

●Battery System Response Data:

CAN ID：0x7310+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x731 (11 bits identifier)

Hardware Version: such as V2.1

Software Version: such as V1.2

Byte0	Reserve (Hardware Version)	0: Null; 1: ver.A; 2: ver.B; Others: Reserve.
Byte1	reserve	
Byte2	Hardware Version-V	0x02
Byte3	Hardware Version-R	0x01
Byte4	Software Version-V (Major)	0x01
Byte5	Software Version-V (Minor)	0x02
Byte6	Reserve (Software Version (Major))	
Byte7	Reserve (Software Version (Minor))	

CAN ID：0x7320+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x732 (11 bits identifier)

Byte0	Battery Module Qty.	
Byte1		
Byte2	Battery Module in series Qty.	
Byte3	Cell Qty. in battery module	
Byte4	Voltage Level	Resolution: 1V Offset: 0
Byte5		
Byte6	AH number	Resolution: 1AH Offset: 0
Byte7		

CAN ID：0x7330+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x733 (11 bits identifier)

Byte0	Manufacturer Name	'S'
Byte1	Manufacturer Name	'E'
Byte2	Manufacturer Name	'N'
Byte3	Manufacturer Name	'E'
Byte4	Manufacturer Name	'R'
Byte5	Manufacturer Name	'G'
Byte6	Manufacturer Name	'Y'
Byte7	Manufacturer Name	0

CAN ID：0x7340+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x734 (11 bits identifier)

Byte0	Manufacturer Name	0
Byte1	Manufacturer Name	0
Byte2	Manufacturer Name	0
Byte3	Manufacturer Name	0
Byte4	Manufacturer Name	0
Byte5	Manufacturer Name	0
Byte6	Manufacturer Name	0
Byte7	Manufacturer Name	0

2 Control Command

2.1 Sleep/Awake Command

Broadcast not supported

Host device send data:

CAN ID：0x8200+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x620 (11 bits identifier)

Byte0	Sleep Awake Control	0x55: Control device enter sleep status; 0xAA: Control device quit sleep status; Others: Null
Byte1	Reserve	0
Byte2	Reserve	0
Byte3	Reserve	0
Byte4	Reserve	0
Byte5	Reserve	0
Byte6	Reserve	0
Byte7	Reserve	0

No response from battery.

2.2 Charge/Discharge Command

Broadcast not supported

Host device send data:

CAN ID：0x8210+Addr(29 bits identifier) (Equipment Address: Addr=0~F, MAX 15 equipments can be parallel communicated.)

CAN ID: 0x621 (11 bits identifier)

Byte0	Charge Command	0xAA: effect; Others: Null (* Note 1)
Byte1	Discharge Command	0xAA: effect; Others: Null (* Note 2)
Byte2	Reserve	
Byte3	Reserve	
Byte4	Reserve	
Byte5	Reserve	
Byte6	Reserve	
Byte7	Reserve	

No response from battery.

*Note:

1. Charge Command: When the battery is in under-voltage protection, the relay is open. When EMS or PCS is going to charge the battery, send this command, then the battery will close the main relay. If the battery is in sleep status, wake up first then use this command.
2. Discharge Command: When the battery is in over-voltage protection, the relay is open.When EMS or PCS is going to discharge the battery, send this command, then the battery will close the main relay. If the battery is in sleep status, wake up first then use this command.